ZCAS University

CCS3232 – SOFTWARE DESIGN AND ARCHITECTURE

FINAL / END OF SEMESTER EXAMINATION

Monday 27th November, 2023

16:30 HRS-19:30HRS

TIME ALLOWED: **WRITING – THREE HOURS**

READING TIME – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

Software Design and Architecture focus on translating the requirements gathered during the systems analysis phase into a well-structured, modular, and efficient software solution. This involves the decomposition of a system into different components and their interactions to satisfy functional and non-functional requirements. Various visual representations (diagrams), tools and frameworks are used to achieve this goal. They include but not limited to the following:

i) Unified Modeling Language (UML)

a) You are required to provide a brief description to explain each of the following UML Diagrams:

- Class Diagram
- Component Diagram
- Deployment Diagram **(9 Marks)**

b) In relation to a Human Resource Management System, you are required to identify and discuss ANY two features / components associated with EACH of the above UML Diagrams. [Clearly state the assumptions you have made, if any] **(12 Marks)**

ii) Architecture View Model (4+1 view model)

a) Identify the four (4) main Architecture View Models and briefly explain the role associated with each view model. **(12 Marks)**

b) Explain why the Architecture View Model is sometimes referred to as "4+1 View Model" instead of "5 View Model". **(3 Marks)**

iii) ADL (Architecture Description Language)

UML (Unified Modeling Language) and ADL (Architecture Description Language) are both used in software design and architecture. However, they serve different purposes and have distinct characteristics. You are required to briefly describe the main differences between UML and ADL. **(4 Marks)**

(Total: 40 Marks)

SECTION B: Attempt any THREE questions in this section

Question 2

Modularity is the concept of dividing a system into smaller, self-contained units or modules. Each module represents a distinct and cohesive piece of functionality within the system. You are required to:

i) Identify and state any four (4) benefits that are associated with modularity. **(4 Marks)**

ii) Provide a brief description for each of the four (4) identified benefits above; clearly stating the associated benefit for each. **(12 Marks)**

iii) Explain the difference between concept of "Modularity" and that of the "Separation of Concerns". **(4 Marks)**

(Total: 20 Marks)

Question 3

Emerging trends in software design and architecture represent the latest developments and shifts in how software systems are conceived, developed, and maintained. These trends are influenced by technological advancements, changing industry/user needs, evolving best practices and the growing complexity of software systems. The following concepts are some of these emerging trends in software design and architecture: Microservices Architecture, Serverless Computing, Containerization, Security-First Design, and Low-Code Development. You are required to write brief notes to explain each of the stated five (5) concepts. **(Total: 20 Marks)**

Question 4

User Interface Architecture: User interface is the first impression of a software system from the user's point of view. It plays a crucial role in any software system. User interface mainly performs two functions: Accepting the user's input and displaying the output.

i) There are basically two major types of user interface: Command-Line Interface, and Graphical User Interface - GUI. Using an appropriate example for each, you are required to explain the difference between the two User Interfaces. **(4 Marks)**

ii) A Graphical User Interface is the most common type of user interface available today. It is also usually referred to as a WIMP interface. The initials WIMP represents four (4) distinct features that are associated with user interface. You are required to identify and briefly explain each of these features. **(8 Marks)**

iii) You can interface with a computer in many varied ways – not just typing on a keyboard or clicking using a mouse. Voice recognition (in both directions) is now becoming established – probably the most natural way for humans to interact. Touch is another very natural human sense – hence the development of touch-screen technology.

a) Using an appropriate example, you are required to explain how voice recognition may be used as a form of a user interface architecture. **(4 Marks)**

b) Using an appropriate example, you are required to explain how the touch-screen technology may be used as a form of a user interface architecture. **(4 Marks)**

(Total: 20 Marks)

Question 5

A Student Management System is a software application or information system designed to efficiently and effectively manage various aspects of student-related data, processes, and interactions within an educational institution. It serves as a centralized platform to access and manage student information, academic records, and related activities.

i) In Software Development Processes, system requirements are normally identified at an early stage. Explain and state the difference between functional and non-functional requirements of a computer system. **(4 Marks)**

ii) In determining system requirements, explain why domain knowledge is considered to be an important element. **(4 Marks)**

iii) Identify and briefly explain any three (3) functional requirements that are associated with a Student Management System. **(6 Marks)**

iv) Identify and briefly explain any three (3) non-functional requirements that are associated with a Student Management System. **(6 Marks)**

(Total: 20 Marks)

END OF EXAMINATION